

Homework - Conditional Statements

- 1) Write a program in a script file that determines the real roots of a quadratic equation. Name the file quad roots. When the file runs, it asks the user to enter the values of the constants a, b, and c. To calculate the roots of the equation the program calculates the discriminant D, given by:

$$D = b^2 - 4ac$$
$$x_{1,2} = \frac{(-b \pm \sqrt{D})}{2a}$$

- a) If $D > 0$, the program displays message “The equation has two roots,” and the roots are displayed in the next line.
- b) If $D = 0$, the program displays message “The equation has one root,” and the root is displayed in the next line.
- c) If $D < 0$, the program displays message “The equation has no real roots.”

Note: let us denote the complex roots as well, how can the conditional structure be written without using a comparison operator?

- 2) The following are formulas for calculating the training heart rate (THR) for men and women.
- a) For men: $THR = [(220 - AGE) - RHR]INTEN + RHR$
 - b) For women: $THR = [(206 - 0.88AGE) - RHR]INTEN + RHR$

where AGE is the person’s age, RHR the resting heart rate, and INTEN the fitness level (0.55 for low, 0.65 for medium, and 0.8 for high fitness). Write a program in a script file that determines the THR. The program asks users to enter their gender (male or female), age (number), resting heart rate (number), and fitness level (low, medium, high). The program then displays the training heart rate.

- 3) Write a program in a script file that calculates the cost of a telephone call according to the following price schedule:

Time the call made	Duration of call		
	1–10 min	10–30 min	More than 30 min
Day: 8 A.M. to 6 P.M.	\$0.10/min	\$1.00 + \$0.08/min for additional min above 10.	\$2.60 + \$0.06/min for additional min above 30.
Evening: 6 P.M. to 12 A.M.	\$0.07/min	\$0.70 + \$0.05/min for additional min above 10.	\$1.70 + \$0.04/min for additional min above 30.
Night: 12 A.M. to 8 A.M.	\$0.04/min	\$0.40 + \$0.03/min for additional min above 10.	\$1.00 + \$0.02/min for additional min above 13.

The program asks the user to enter the time the call is made (day, evening, or night) and the duration of the call (a number that can have one digit to the right of the decimal point). If the duration of the

call is not an integer, the program rounds up the duration to the next integer. The program then displays the cost of the call.

- 4) Write a program in a script file that converts a measure of volume given in units of either m³, L, ft³, or gal (U.S. gallons) to the equivalent quantity in different units specified by the user. The program asks the user to enter the amount of volume, its current units, and the desired new units. The output is the specification of volume in the new units. Use the program to:
 - a) Convert 3.5 m³ to gal.
 - b) Convert 200 L to ft³.
 - c) Convert 480 ft³ to m³.
- 5) The overall grade in a course is determined from the grades of 5 quizzes, 3 midterms, and a final, using the following scheme: Quizzes: Quizzes are graded on a scale from 0 to 10. The grade of the lowest quiz is dropped and the average of the 4 quizzes with the higher grades constitutes 25% of the course grade. Midterms: Midterms are graded on a scale from 0 to 100. If the average of the midterm scores is higher than the score on the final, the average of the midterms is 35% of the course grade. If the final grade is higher than the average of the midterms, then the lowest midterm is dropped and the average of the two midterms with the higher grades is 35% of the course grade. Final: Finals are graded on a scale from 0 to 10. The final is 40% of the course grade. Write a computer program in a script file that determines the course grade for a student. The program first asks the user to enter the five quiz grades (in a vector), the three midterm grades (in a vector), and the grade of the final. Then the program calculates a numerical course grade (a number between 0 and 100). Finally, the program assigns a letter grade according to the following key: A for, B for, C for, D for, and E for a grade lower than 60. Execute the program for the following cases:
 - a) Quiz grades: 7, 9, 4, 8, 7. Midterm grades: 93, 83, 87. Final grade: 89.
 - b) Quiz grades: 8, 6, 9, 6, 9. Midterm grades: 81, 75, 79. Final grade: 72.